Statement of Purpose

Daman Arora
daman1209arora@gmail.com

The strongest modern general-purpose AI systems are unable to reason faithfully [Valmeekam et al., 2023, Arora et al., 2023a]. My research goal is to build general purpose AI models which are capable of reasoning and planning.

Learning to reason is hard. I believe there are multiple factors which contribute to this:

1. "Predicting the next token from a corpus of text allows Language Models to learn a world model". **But which world model?** This corpus is just a **projection** of the world model. Can neural networks discover the real "Chain-of-Thoughts" from learning through a supervised dataset? Research shows that neural methods can fail to discover the true generative process Zhang et al. [2022] behind reasoning.
2. The capabilities of transformer based models seem to be limited, exhibiting a variability in capacities of learning different types of algorithms/structures [Dziri et al., 2023, Zhou et al., 2023]. What would be an architecture which allows true **generalization** to tasks requiring exact computations?
3. LLMs have not been taught to explore. **Exploration** forms the basis of mathematical creativity and is necessary for a lot of tasks such as planning and proof generation. A feedback mechanism to LLMs from self-verification along with guided exploration is necessary in order to make them strong and creative reasoners which could, one day, help humans in research.

My interest in this field has been a result of my past research experience in varied subjects, which I describe in the following section. I believe that my previous learnings will help me in my future endeavours.

Research in Generalized Planning: My journey with ML research started during my third year of study at IIT Delhi with a project on Generalized Planning with Dr. **Mausam** and Dr. **Parag Singla**. The project's goal was: given small training instances of a relational MDP from a particular domain, generalize to larger instances. A relational MDP is a first-order representation of a class of MDPs with varying number of objects, but a shared transition and reward function. We focused on learning a generalized policy which would give good rewards for all instantiations of the RMDP. The SymNet architecture, introduced previously, consisted of learning a GNN (Graph Neural Network) based policy on an instance graph, which was a graph representation of an MDP. Since the policy was parameterized by a GNN, it was size invariant.

When I began working on SymNet, I noticed that training using Reinforcement Learning was unstable which made it hard to make any improvements to the underlying architecture. Pivoting to a more stable Imitation Learning approach unearthed a treasure trove of tractable problems. **Representational inadequacy** of the SymNet graph formulation was clear from our experiments. We decided to fix these by creating a new method of converting a relational MDP to a graph which alleviated the lack of representability in SymNet, and we came with the second version. Our experiments surprisingly revealed that **SymNet2.0**, unlike SymNet was able to beat its teacher on larger instances. This work got published in **UAI 2022** [Sharma et al., 2022].

However, SymNet2.0's architecture was severely limited by the depth of the GNN used for learning. However, there existed a large class of problems which required **"long-range dependencies"**. The notion of distance in a relational MDP wasn't well defined at that time. Coming up with a useful definition using **Dynamic Bayesian Networks**, we decided to implement a novel GNN architecture called **SymNet3.0** which could incorporate these distances using intermediate GNN layers having all-pair attention using distances as edge features. In addition to our empirically grounded work, I was able to get **theoretical guarantees** on SymNet3.0 which added to its utility. This work got published at **UAI 2023**, [Sharma et al., 2023] with me as a co-first author.

Research in Reasoning and Planning using LLMs: After my stint with Generalized planning, I decided to diversify. The field of "generative AI" was booming with GPT-3.5 coming out to the general public, and its inability to plan piqued my interest. I collaborated with Dr. **Subbarao Kambhampati** from ASU on this subject. I was curious if LLMs could even simulate a PDDL domain, let alone plan. My intuition was that maybe LLMs can plan if fine-tuned, unlike previous works which had tried prompting LLMs to generate plans. My experience with a small LM (GPT-2) revealed that even with fine-tuning, LMs hallucinate and generate invalid actions. How could we improve plans without resorting to external feedback? This was a question which bothered me and for this, I came up with a simple solution of learning **verifiers** from the same fine-tuning data on which the plan generation is trained by sampling negative data by choosing random actions. Surprisingly, this simple idea worked very well in practice, and we published a workshop paper [Arora and Kambhampati, 2023] in the **ICML Workshop on Knowledge and Logical Reasoning in the Era of Data-driven Learning**.

Concurrently, I was also interested in more 'tacit' problem solving skills such as those present in mathematics and engineering. The JEE Advanced exam seemed like a perfect test-bench to evaluate GPT-4. We were pleasantly surprised by our initial results which showed that GPT-4's performance was highly non-trivial. I decided to take the **initiative** to consolidate this research and approached Dr. Mausam with the research proposal. We curated the **JEEBench** dataset consisting of questions from past 8 years' question papers and evaluated various LLMs on the dataset. I believed that more than the dataset, insights would be far more important for the research community. Significant human evaluation efforts were made to mine actionable insights, both qualitative and quantitative. We worked on a simple method to make the best out of an LLM under uncertainty posed by thresholding on confidence levels generated using self-consistency responses. In our work, we were probably the first to demonstrate that the "too-good-to-be-true" idea that LLMs can self-improve does not work. A lot of works in the near future came up with similar results! Our work got accepted at **EMNLP 2023** [Arora et al., 2023a] and received generous reviews from reviewers and was covered by a national news paper (Hindustan Times).

Research in Information Retrieval: At Microsoft Research, I'm currently working on an experimental approach of utilizing feedback from a retrieval system's result to improve retrieval quality. In our recent work, with my mentor **Nagarajan Natarajan** we proposed a approach which merges GAR (Generation Augmented Retrieval) and RAG (Retrieval Augmented Generation) [Arora et al., 2023b]. We utilize an LLM as a meta-controller of a retriever which attempts to rewrite queries in order to adapt it to the retriever's strengths.

Future research goals: I find myself deeply intrigued by these areas of research and the PhD program at CMU would help me in exploring these hard problems. I want to (i) build a strong theoretical basis of the limits of machines which reason via learning, (ii) find better learners than Transformer based architectures, (iii) integrate curiosity and exploration into the fundamental learning cycle of LLMs and finally, (iv) build AI for collaborating with humans for research. My experience with generalized planning and planning using LLMs has given me important insights into the limits of reasoning via learning. Additionally, working on the JEEBench dataset has given me great intuition of the state-of-the-art of ML.

At CMU, I would like to work with Prof. **Yuanzhi Li**, Prof. **Aviral Kumar** and Prof. **Albert Gu**. I am inspired by Prof. Li's work on building small specialized models like Phi-1.5 [Li et al., 2023], the study of generalization properties in transformer type models [Jelassi et al., 2023] and the theoretical study of generalization in RL [Malik et al., 2021]. Prof Kumar's work on offline RL [Kumar et al., 2022] and generalization in RL [Ghosh et al., 2021] is something I have followed and employed in my work on generalized planning. Prof. Gu's work on Mamba [Gu and Dao, 2023] is very exciting because of the architecture's improved performance and higher theoretical representability (Transformers v. RNNs).

I hope my PhD program would be the most satisfying and fruitful endeavor of my career. Research gets me the most excited. I believe that exploratory research is thrilling and has transformatory potential. After my PhD, I wish to stay in academia since I believe universities have a more open-ended research agenda which allows more space for creativity. I am also excited about mentoring and teaching other students which I personally feel is invaluable.

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